

LeverageSHAP — Method Summary

C. Musco & R. T. Witter, *Provably Accurate Shapley Value Estimation via Leverage Score Sampling*, arXiv:2410.01917

1. The Estimator

- Leverage SHAP: modification of the widely used Kernel SHAP algorithm
- Both are based on the same fact: Shapley values are the solution of a weighted least-squares regression over all $2^n - 2$ coalitions
- Strict rule attached: all values combined must exactly equal the total output of the model (**efficiency axiom**): $\sum_i \varphi_i = v(\mathbf{1}) - v(\mathbf{0})$
- Important: the constraint itself is NOT the problem — Kernel SHAP solves it fine via Lagrange multipliers
- The real reason for the reformulation: **leverage-score theory only exists for unconstrained regression** $\Rightarrow$ the constraint has to disappear first
- The trick is geometric: split the solution into a known and an unknown part

$$\varphi = \varphi^\perp + \delta \mathbf{1}, \quad \delta = \frac{v(\mathbf{1}) - v(\mathbf{0})}{n}$$

- $\delta \mathbf{1}$ = uniform “fair share” per feature $\Rightarrow$ known before any regression
- $\varphi^\perp$ (entries sum to 0) = what makes the features *differ* $\Rightarrow$ the only thing the regression must find
- To stop the solver from leaving the zero-sum subspace: filter Z with a projection, $A = ZP$, $P = I - \frac{1}{n} \mathbf{1}\mathbf{1}^\top$ — in practice just row-centering ($z \mapsto z - \frac{s}{n}$, $s = \|z\|_1$)
- Target shifted consistently: $b_z = (v(z) - v(\mathbf{0})) - \delta s$
- Solve unconstrained least squares for $\varphi^\perp$, then add $\delta \mathbf{1}$ back $\Rightarrow$ efficiency holds *exactly*, no constrained solver needed
- Nice side effect: $A^\top A$ is a scaled projection $\Rightarrow$ all non-zero singular values equal $\Rightarrow$ perfectly conditioned, numerically stable even for $n \approx 100$

2. The Sampling Scheme

- Kernel SHAP picks coalitions by a heuristic kernel; Leverage SHAP uses **leverage scores** = the principled measure of how much a row influences a regression
- Normally intractable for 2^n rows — but Lemma 3.2 proves they only depend on coalition size:

$$\ell_z = \binom{n}{\|z\|_1}^{-1}$$

- $\Rightarrow$ every 3-feature coalition has the exact same leverage as any other 3-feature coalition; every *size* gets equal total probability (Kernel SHAP over-weights the extreme sizes)
- **Paired sampling**: whenever z is drawn, its complement $\bar{z}$ is drawn too (valid since $\ell_z = \ell_{\bar{z}}$)
- **Bernoulli sampling without replacement**: each pair included independently with prob. $\min(1, 2c \cdot \ell_z)$; c from binary search so the expected budget fits:

$$m - 2 = \sum_{s=1}^{n-1} \min\left(\binom{n}{s}, 2c\right)$$

- $\Rightarrow$ sizes with $\binom{n}{s} \leq 2c$ get enumerated *completely* — which is why the estimator becomes exact as $m \rightarrow 2^n$; large sizes: per-size Binomial draw + lexicographic unranking
- **IS reweighting**: we sampled by leverage but the regression needs the Shapley kernel $\Rightarrow$ reweight each row by $w(s) / \min(1, 2c \cdot \ell_z)$
- In the subsampled regime the huge $\binom{n}{s}$ (up to $\sim 10^{29}$ for $n = 100$) cancels *analytically*, leaving bounded weights $\propto 1/(s(n-s))$ — by design, not luck; fully enumerated sizes just keep $w(s)$

3. Claimed Convergence

- Kernel SHAP: no non-asymptotic guarantee exists for any $m < 2^n - 2$
- Leverage SHAP provides exactly this: with $m = O(n \log(n/\delta) + n/(\varepsilon\delta))$ evaluations, the regression objective is within a $(1 + \varepsilon)$ factor of optimal, with prob. $1 - \delta$ (Theorem 1.1) $\Rightarrow$ nearly linear budget, not exponential
- Perfect conditioning translates this into an ℓ_2 bound: $\|\varphi - \tilde{\varphi}\|^2 \leq \varepsilon\gamma\|\varphi\|^2$, with γ = how far the game is from linear (appendix confirms: error really grows with γ , for all regression methods)
- Why only objective error? Per-value multiplicative approximation is NP-hard $\Rightarrow$ this is the strongest guarantee one can reasonably ask for

4. The Experiments

- 8 real-world datasets from the SHAP library ($n = 4$ up to 101), XGBoost models
- Ground truth via Tree SHAP (exact Shapley values computable for tree models); metric: relative ℓ_2 error; budgets $m = 5n \dots 160n$, 100 runs, medians + quartiles
- Benchmarked against standard Kernel SHAP and the highly optimized SHAP-library version
- Result: Leverage SHAP wins at every budget — typically its *third quartile* sits at the *median* error of Optimized Kernel SHAP
- Advantage especially massive for large n and small budget m — exactly the regime that matters in practice
- Extras to reproduce: machine precision once $m = 2^n$ (small n); best under noisy value functions; ablations: pairing helps everywhere, Bernoulli sampling mainly for small n